

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		Activation energy	1			1		
		(ii)		Curve drawn with a lower activation energy under existing curve – energy state at beginning and end must be the same.	1			1		
	(b)			{Alcohol dehydrogenase/ enzyme} has a <u>specific</u> {shaped active site/ tertiary structure/ OWTTE} (1) {Ethanol/ substrate} has a <u>complementary</u> (shape) (1) (The two fit together) to form an <u>enzyme-substrate complex</u> (1)	3			3		
	(c)	(i)		Any answer between 0.57 - 0.63 = 2 marks If incorrect award 1 mark for sight of : attempted calculation of gradient		2		2	2	
		(ii)		1. P: the rate of reaction is higher as the concentration of {ethanol/ substrate} is high (1) 2. The concentration of {alcohol dehydrogenase/ enzyme} is limiting the rate of reaction; (1) 3. Q: the rate of reaction is lower the concentration of {ethanol/ substrate} decreases(1) 4. The concentration of {ethanol/ substrate} becomes the limiting factor. (1)		2	2	4		

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6	(d)			1. Ethanol and ethylene glycol must have a similar structure / Ethanol and ethylene glycol must both be complementary to the shape of the active site of {alcohol dehydrogenase/ the enzyme} (1) 2. ethanol acts as a competitive inhibitor (1) 3. When ethanol binds to the active site it prevents ethylene glycol from attaching (1) 4. Fewer enzyme-substrate complexes form (1) 5. Which reduces the rate of production of {glycoaldehyde/ product} (1)		3	2	5		
				Question 6 total	5	7	4	16	2	